BusYatra

Online Bus Ticket Booking System

A Cornerstone Project Report

Submitted in partial fulfillment of the requirements for the

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Department of Computer Science & Engineering

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SCHOOL OF ENGINEERING

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Cornerstone project work entitled

"Online Bus Ticket Booking System (BusYatra)"

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ABSTRACT

The BusYatra – Online Bus Ticket Booking System is a desktop-based application developed to provide a seamless and efficient platform for booking bus tickets. The system allows users to search buses, check seat availability, book tickets, cancel bookings, and manage their travel records.

The application includes administrative features such as bus management, route scheduling, booking monitoring, and user management. It reduces manual effort involved in traditional booking systems and ensures accuracy, speed, and reliability.

The system is developed using Java Swing for GUI, MySQL for database management, and JDBC for database connectivity. It provides a secure, user-friendly, and scalable solution for managing bus ticket reservations effectively.

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ACRONYMS

Acronym	Full Form
JDBC	Java Database Connectivity
GUI	Graphical User Interface
SQL	Structured Query Language
IDE	Integrated Development Environment
JVM	Java Virtual Machine
OOP	Object-Oriented Programming
SSL	Secure Sockets Layer
DAO	Data Access Object
DBMS	Database Management System
API	Application Programming Interface

CHAPTER 1 – INTRODUCTION

1.1 Brief Information about the Project

The Bus Ticket Booking System (BusYatra) is a desktop-based system that provides an efficient solution where users can book tickets directly from their system, check seat availability, and view bus schedules instantly. The system automates ticket booking, cancellation, and record management, making the entire process faster and more reliable. It also helps administrators easily manage data and monitor bookings without manual effort.

The system replaces traditional manual booking methods with a computerized solution, thereby improving data accuracy, reducing redundancy, and enhancing accessibility. The application is developed using Core Java, Swing (GUI), and JDBC (database connectivity). The system uses a MySQL database hosted on the Aiven cloud platform for storing user, bus, and booking information securely.

1.2 Motivation and Contribution of Project

Traditional bus booking systems faced the following challenges:

- Long queues at ticket counters
- Manual errors in record keeping
- Poor record management and accessibility

BusYatra solves these issues by:

- **Providing online-style booking in a desktop application**
- **Automating seat allocation and availability tracking**
- **Reducing paperwork and human errors**
- **Improving speed, accuracy, and overall efficiency**

1.3 Objectives of the Project

The main objectives of the Online Bus Ticket Booking System are:

- To develop a user-friendly application for booking bus tickets
- To store user, bus, and booking information in a structured database

- To provide functionalities for booking, cancelling, and managing tickets
- To enable quick searching of bus schedules based on source and destination
- To display booking and bus details in an organized format
- To ensure accurate seat availability and efficient ticket management

1.4 Scope of the Project

The Online Bus Ticket Booking System (BusYatra) can be used by passengers and administrators to manage bus ticket reservations efficiently. It supports operations such as user registration, viewing bus schedules, checking seat availability, booking tickets, cancelling bookings, and managing bus details.

The system provides a simple and user-friendly interface for users to book tickets and for administrators to maintain bus and booking information in a centralized database. It ensures quick access to travel details and helps in reducing manual work, making the overall booking process more efficient and reliable.

CHAPTER 2 – SYSTEM ANALYSIS

2.1 Functional Requirements

Functional requirements define the core operations that the Online Bus Ticket Booking System must perform to effectively manage ticket booking and related information.

User Registration and Login

The system should allow users to register and log in using their credentials. Proper validation should be applied to ensure secure access.

View Bus Details

The system should provide functionality to view available bus information including routes, timings, and seat availability.

Ticket Booking

The system should allow users to book bus tickets by selecting the desired bus and seat. It should update seat availability automatically after each booking.

Ticket Cancellation

The system should provide functionality to cancel booked tickets. The seat should be made available again after cancellation.

Search Bus

The system should support searching for buses based on source and destination. The search results should be accurate and quick.

Manage Bus Details (Admin Module)

The system should allow the administrator to add, update, and delete bus details such as routes, timings, and seat capacity.

View Booking Records

The system should display booking details in an organized format for both users and administrators.

2.2 Non-Functional Requirements

Performance

The system should respond quickly to operations like search, book, and retrieve without delays.

Security

Only authorized users should access and modify data using proper authentication and access control.

Usability

The system should be simple and easy to use, allowing users to perform tasks without difficulty.

Reliability

The system should operate consistently and handle errors without data loss.

Scalability

The system should handle increasing data and users without affecting performance.

2.3 Requirements Specification

2.3.1 Minimum Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB (minimum)
- Hard Disk: 20 GB free space
- Monitor: Standard display
- Keyboard and Mouse

2.3.2 Software Requirements

- **Operating System: Windows 7 / 10 / 11**
- **Programming Language: Java**

- **IDE: NetBeans**
- **Database: MySQL (Aiven Cloud)**
- **Connectivity: JDBC**

CHAPTER 3 – TECHNOLOGY DESCRIPTION

3.1 Programming Language

3.1.1 Introduction to Java

Java is a high-level, object-oriented programming language developed by Sun Microsystems. It is widely used for developing desktop, web, and enterprise applications due to its platform independence and robustness.

In the BusYatra – Online Bus Ticket Booking System, Java is used to implement the core functionality of the application. It handles operations such as user authentication, bus search, seat selection, ticket booking, cancellation, and data processing. Java provides features like object-oriented programming, exception handling, multithreading, and strong memory management.

3.2 Framework / Libraries

3.2.1 Introduction to Java Swing

Swing is a part of Java that is used to create graphical user interfaces (GUI). It provides a set of components such as buttons, labels, text fields, tables, and frames to design user-friendly applications.

In this project, Swing is used to design the following screens of BusYatra:

- Login Window
- Dashboard / Main Menu
- Bus Search Screen
- Seat Selection Interface
- Booking Confirmation Screen
- Cancellation Panel
- Admin Dashboard

3.3 Database Technology

The BusYatra System uses a relational database for storing and managing data such as users, buses, bookings, and seat availability in a structured manner. This project uses a cloud-based MySQL database hosted on Aiven, which provides managed database services ensuring better scalability, reliability, and remote accessibility.

Key advantages of using Aiven Cloud Database:

- Remote access from anywhere
- High availability and uptime
- Secure connections with SSL support
- Automatic backups and maintenance
- Scalable infrastructure

3.3.1 JDBC (Java Database Connectivity)

JDBC (Java Database Connectivity) is an API used to connect Java applications with databases. It allows execution of SQL queries and retrieval or updating of data. In BusYatra, JDBC connects the Java application to the MySQL database hosted on Aiven. JDBC performs the following operations:

- Establishing connection with the database
- Executing SQL queries (INSERT, UPDATE, DELETE, SELECT)
- Retrieving data from database tables
- Closing database connections

3.3.2 MySQL Database (Aiven Cloud)

MySQL is an open-source relational database management system used for storing structured data. In the BusYatra system, MySQL (hosted on Aiven cloud platform) is used to store user details, bus information, booking records, seat availability, and transaction data. MySQL ensures data integrity using Primary Keys, Foreign Keys, and NOT NULL constraints.

3.4 Tools & IDEs

3.4.1 NetBeans IDE

NetBeans is used as the development environment for the BusYatra system. It is a powerful and user-friendly Integrated Development Environment (IDE) that provides built-in support for Java development, including code completion, syntax highlighting, built-in GUI builder for Swing, project management, and debugging tools.

CHAPTER 4 – DATABASE DESIGN

4.1 Database Tables

The BusYatra system uses a relational database hosted on Aiven (MySQL) to manage all booking-related operations efficiently. The main tables are:

- Users Table (users) – Stores user account and login details
- Buses Table (buses) – Stores bus information such as operator, type, and seat capacity
- Schedules Table (schedules) – Stores bus schedules (date, timing, routes)
- Bookings Table (bookings) – Stores booking details of users
- Booked Seats Table (booked_seats) – Stores seat allocation for each booking
- Booking Passengers Table (booking_passengers) – Stores passenger details
- Cancellations Table (cancellations) – Stores cancelled ticket details
- Cities Table (cities) – Stores city names used for routes

4.2 Table Structure

4.2.1 Users Table

Field Name	Data Type	Description	Constraint
id	INT	Unique user ID	Primary Key
name	VARCHAR(100)	User name	NOT NULL
email	VARCHAR(100)	Email address	UNIQUE
password	VARCHAR(100)	User password	NOT NULL

4.2.2 Buses Table

Field Name	Data Type	Description	Constraint
id	INT	Bus ID	Primary Key
operator	VARCHAR(100)	Bus operator	–
bus_type	ENUM	AC / NON_AC / SLEEPER / SEATER	–
total_seats	INT	Total seats	NOT NULL
fare_multiplier	DECIMAL	Fare adjustment factor	–

4.2.3 Bookings Table

Field Name	Data Type	Description	Constraint
id	INT	Booking ID	Primary Key
user_id	INT	User reference	Foreign Key
schedule_id	INT	Schedule reference	Foreign Key
from_stop_id	INT	Source stop	Foreign Key
to_stop_id	INT	Destination stop	Foreign Key
total_amount	DECIMAL	Total fare	–
status	ENUM	CONFIRMED / CANCELLED	–
booking_time	TIMESTAMP	Booking timestamp	DEFAULT

4.2.4 Booked Seats Table

Field Name	Data Type	Description	Constraint
id	INT	Unique ID	Primary Key
booking_id	INT	Booking reference	Foreign Key
schedule_id	INT	Schedule reference	Foreign Key
seat_id	INT	Seat reference	Foreign Key

Note: Unique constraint on (schedule_id, seat_id) ensures no duplicate seat booking.

4.2.5 Booking Passengers Table

Field Name	Data Type	Description	Constraint
id	INT	Passenger ID	Primary Key
booking_id	INT	Booking reference	Foreign Key
name	VARCHAR(100)	Passenger name	–
age	INT	Age	–
gender	ENUM	MALE / FEMALE / OTHER	–

4.2.6 Cancellations Table

Field Name	Data Type	Description	Constraint
id	INT	Cancellation ID	Primary Key
ticket_id	INT	Ticket reference	Foreign Key
user_id	INT	User reference	Foreign Key
reason	VARCHAR	Cancellation reason	–
refund_amount	DOUBLE	Refund value	–
status	VARCHAR	Cancellation status	–
cancelled_at	DATETIME	Cancellation time	DEFAULT

4.2.7 Cities Table

Field Name	Data Type	Description	Constraint
id	INT	City ID	Primary Key
name	VARCHAR(100)	City name	UNIQUE

4.3 Keys and Constraints

Keys and constraints are used to maintain data integrity and ensure consistency across all tables.

Primary Keys:

- id column in all tables (users, buses, bookings, schedules, seats, etc.)

Foreign Keys:

- user_id → users
- schedule_id → schedules
- seat_id → seats
- booking_id → bookings

Unique Constraints:

- City name must be unique across the cities table
- Seat cannot be double-booked: (schedule_id + seat_id) is unique

NOT NULL Constraints:

- Critical fields like id, name, and password cannot be null

4.4 Table Relationships

The entity relationships between the database tables are as follows:

Relationship	Type	Description
User → Booking	One-to-Many	One user can have multiple bookings
Booking → Passengers	One-to-Many	One booking can include multiple passengers
Bus → Schedule	One-to-Many	One bus can have multiple schedules
Schedule → Booked Seats	One-to-Many	One schedule can have multiple booked seats
Booking → Cancellation	One-to-One	One booking can be cancelled once

CHAPTER 5 – SYSTEM DESIGN

5.1 System Architecture

The BusYatra – Online Bus Ticket Booking System follows a three-layer architecture to ensure proper separation of concerns, scalability, and maintainability.

1. Presentation Layer (User Interface)

- Developed using Java Swing
- Provides interactive GUI for users and administrators
- Includes: Login Screen, Dashboard, Bus Search, Seat Selection, Booking & Cancellation Panels

2. Application Layer (Business Logic)

- Handles core functionalities of the system
- Processes user inputs: bus search, seat allocation, ticket booking, cancellation
- Implements logic using Java classes and DAO (Data Access Object) pattern

3. Data Layer (Database)

- Stores all system data: Users, Buses, Bookings, Seats, Cancellations
- Managed using MySQL database hosted on Aiven cloud
- Connected to the application layer using JDBC

This layered architecture improves code organization, security, maintainability, and scalability of the overall system.

5.2 Modules Description

1. User Module

- User registration and login with validation
- View available buses based on route
- View complete booking history
- Manage user profile information

2. Bus Management Module (Admin)

- Add new buses to the system
- Update existing bus details
- Delete bus records
- Manage routes, timings, and seat capacity

3. Search Module

- Search buses based on source city, destination city, and travel date
- Displays available buses with details like fare, timing, and available seats

4. Booking Module

- Select preferred bus and seats from the seat layout
- Enter passenger details for each seat
- Confirm ticket booking and generate booking record

5. Cancellation Module

- Cancel a previously booked ticket
- Automatically update seat availability after cancellation
- Calculate and process refund amount
- Maintain cancellation records in the database

6. Seat Management Module

- Displays real-time seat layout of the selected bus
- Tracks available and booked seats
- Prevents double booking using unique constraints

7. Database Connectivity Module

- Establishes connection to Aiven MySQL using JDBC
- Executes all SQL queries (INSERT, UPDATE, DELETE, SELECT)

8. User Interface Module

- Entirely designed using Java Swing components
- Handles all user interactions and navigation between screens

CHAPTER 6 – SYSTEM IMPLEMENTATION

This chapter presents the implementation of the BusYatra system through screenshots of all key screens and modules.

6.1 Login Screen

Displays the login interface where the user enters username and password to access the system.

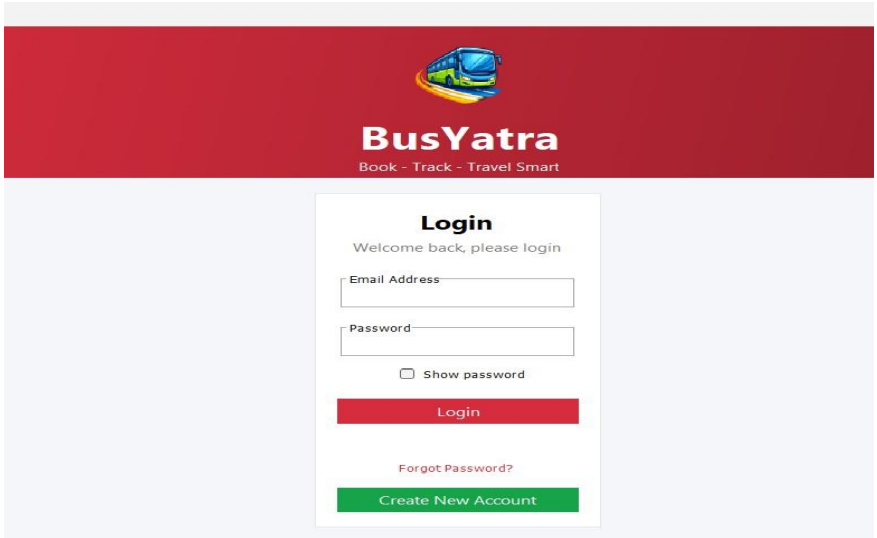

Figure 1: Login Screen of BusYatra System

6.2 Dashboard Screen

The Dashboard provides a centralized interface for users to access all major functionalities including Book Ticket, My Tickets, Wallet, Complaints, and Settings. It also shows wallet balance, booking status, and upcoming trips.

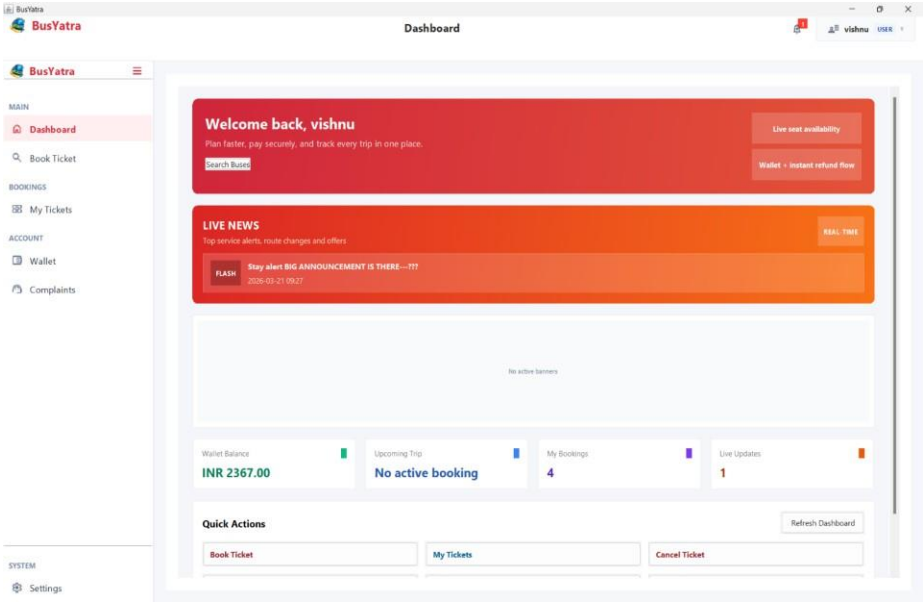

Figure 2: Dashboard Screen of BusYatra System

6.3 Bus Search Screen

Shows the interface where users can search buses based on source city, destination city, and travel date.

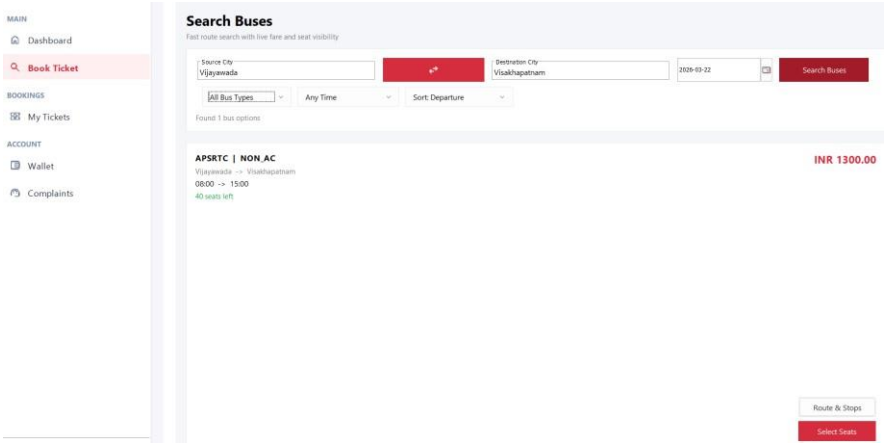

Figure 3: Bus Search Screen of BusYatra System

6.4 Seat Selection Screen

Displays the seat layout of the selected bus and allows users to choose from available seats.

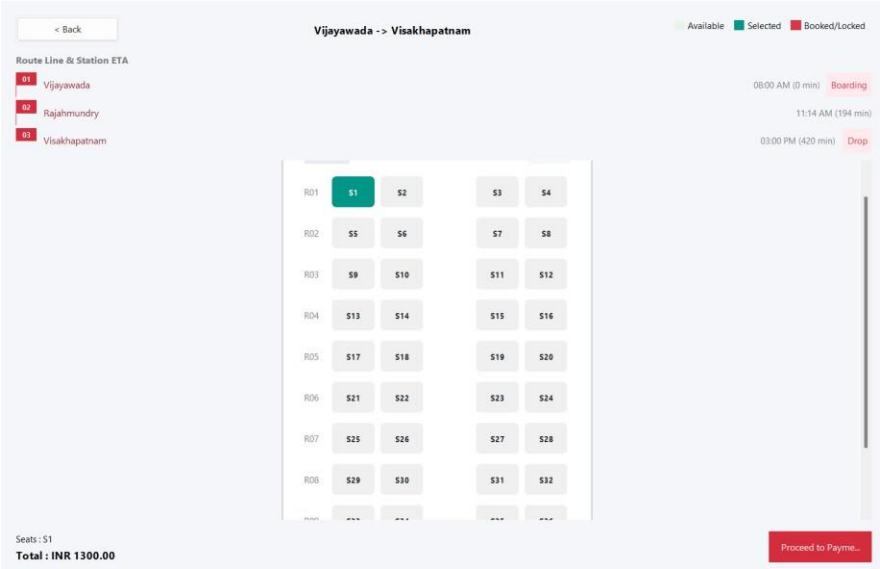

Figure 4: Seat Selection Screen of BusYatra System

6.5 Ticket Booking Screen

Shows the interface for entering passenger details and confirming the ticket booking.

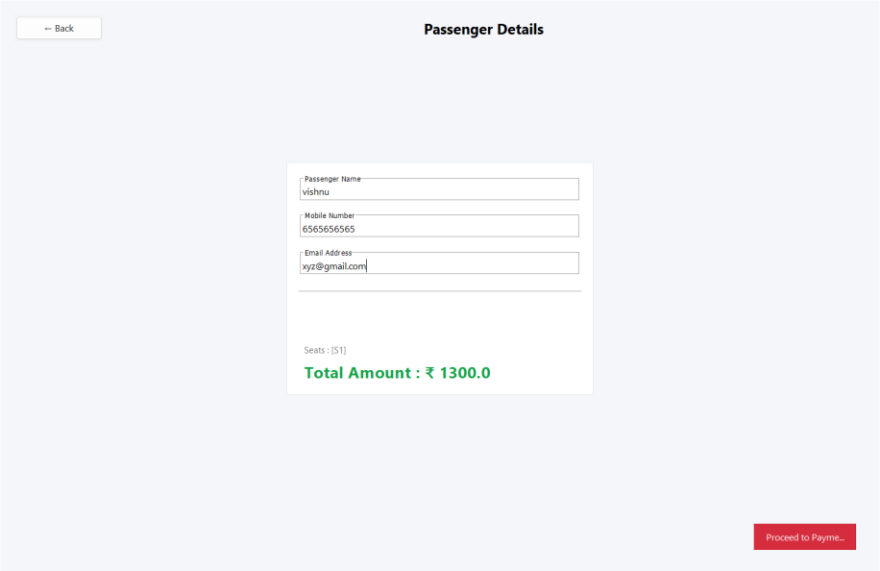

Figure 5: Ticket Booking Screen of BusYatra System

6.6 Payment Screen

Displays the payment interface where users review the booking summary, total fare, and complete the payment.

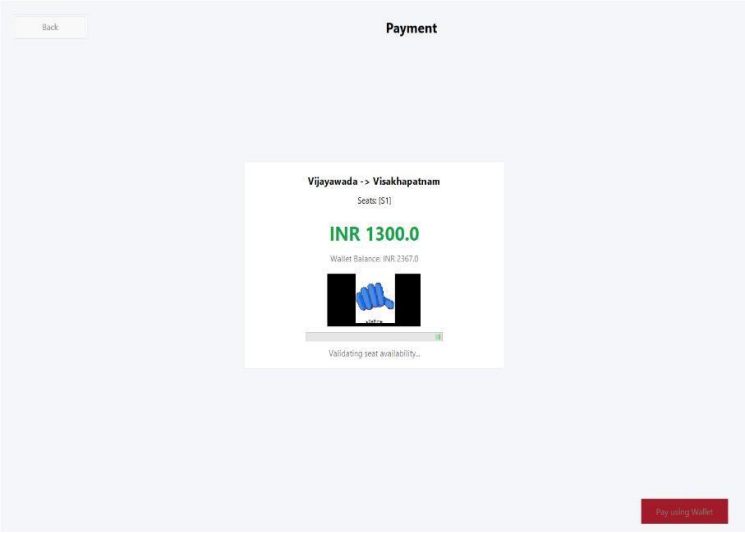

Figure 6: Payment Screen of BusYatra System

6.7 Booking Confirmation Screen

Displays a successful booking confirmation with all important ticket details after payment is completed.

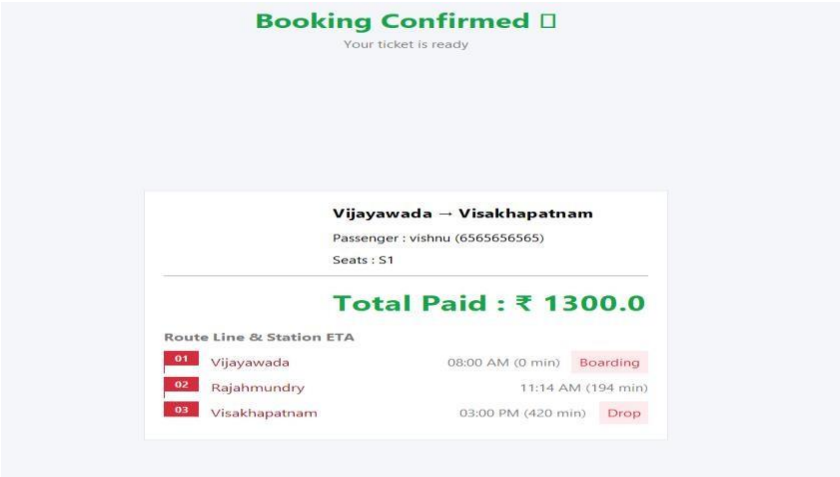

Figure 7: Booking Confirmation Screen of BusYatra System

6.8 Ticket Preview Screen

Displays a preview of the booked ticket showing passenger information, bus details, journey route, date, seat numbers, and total fare.

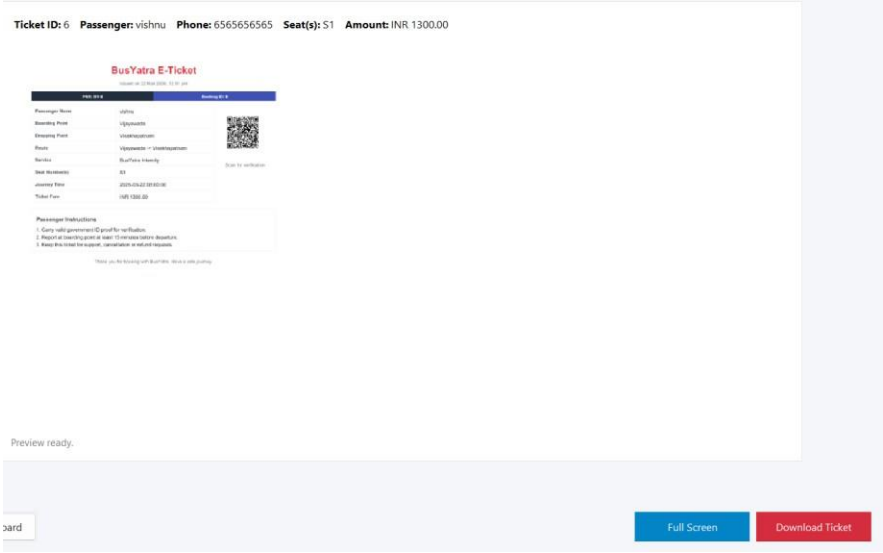

Figure 8: Ticket Preview Screen of BusYatra System

6.9 Ticket Download Screen

Provides the option to download the booked ticket as a PDF for saving or printing.

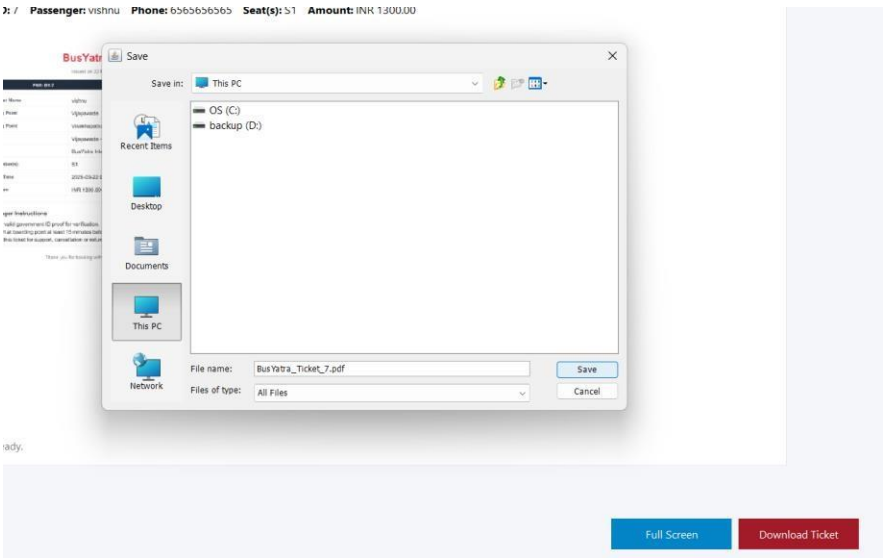

Figure 9: Ticket Download Screen of BusYatra System

6.10 My Tickets Screen

Displays all booked tickets of the logged-in user in a structured tabular format.

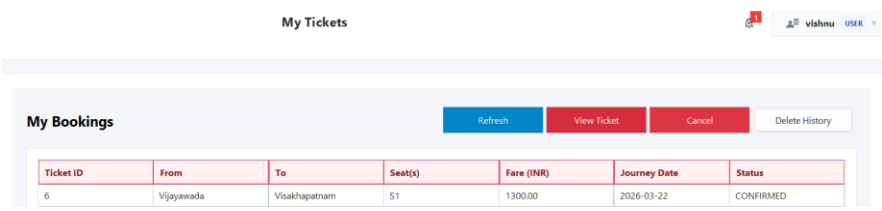

Figure 10: My Tickets Screen of BusYatra System

6.11 Ticket Cancellation Screen

Provides the interface to cancel a booked ticket and process the corresponding refund.

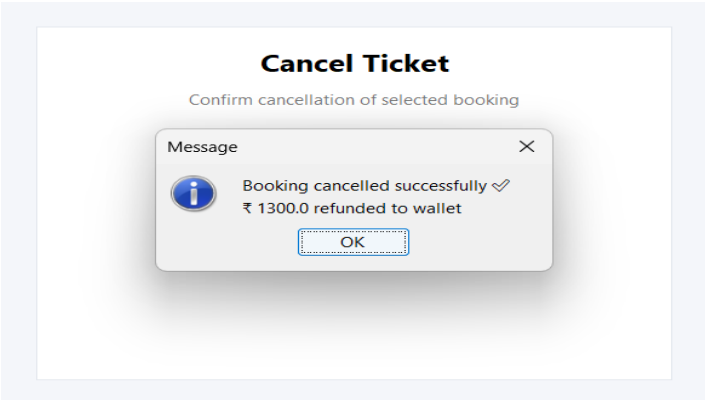

Figure 11: Ticket Cancellation Screen of BusYatra System

6.12 Admin Dashboard Screen

Provides the administrator with an overview of the system and quick navigation to bus, booking, user, and cancellation management modules.

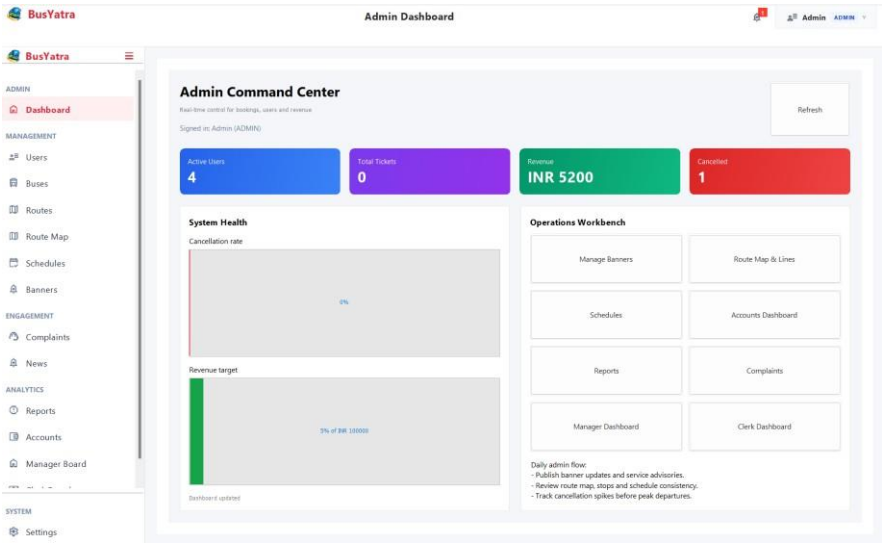

Figure 12: Admin Dashboard Screen of BusYatra System

6.13 Bus Management Screen

Allows the admin to add, update, or delete bus details such as bus number, type, routes, timings, and seat capacity.

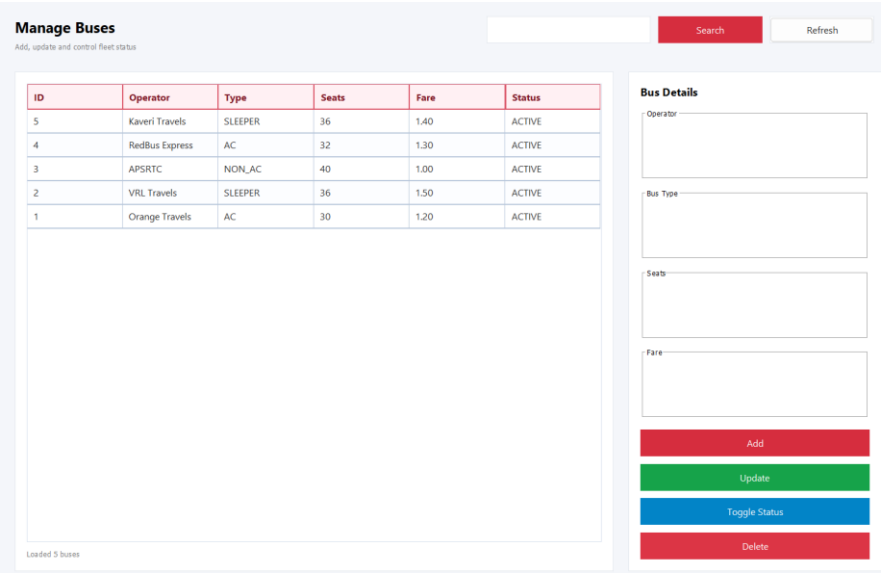

Figure 13: Bus Management Screen of BusYatra System

6.14 Booking Management Screen

Displays all booking records in a structured format, allowing the admin to monitor system activity.

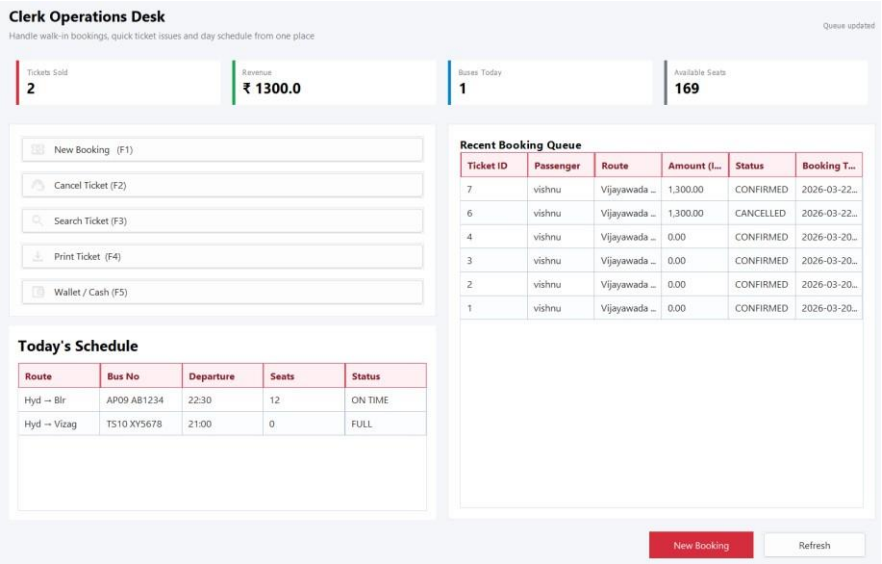

Figure 14: Booking Management Screen of BusYatra System

6.15 User Management Screen

Shows all registered users and allows the admin to view and manage user information.

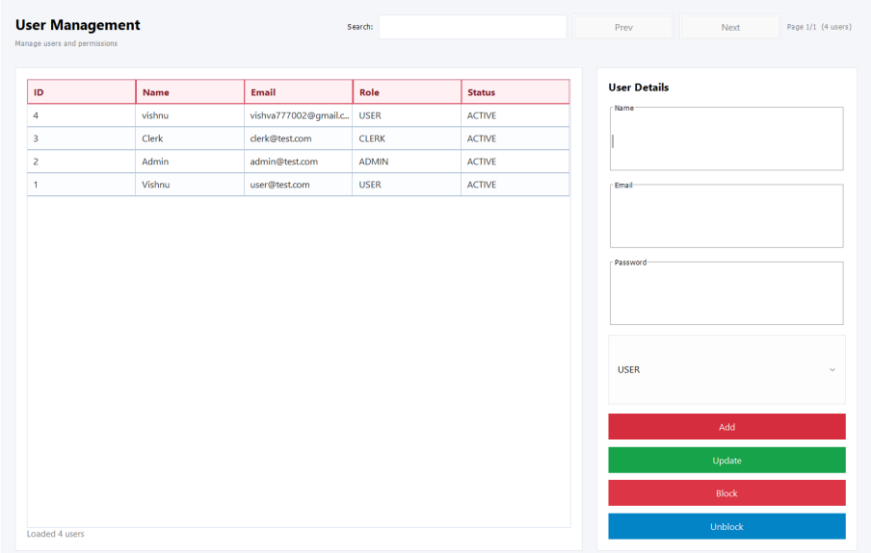

Figure 15: User Management Screen of BusYatra System

6.16 Cancellation Management Screen

Displays cancelled tickets and refund details, allowing the admin to track and manage cancellations.

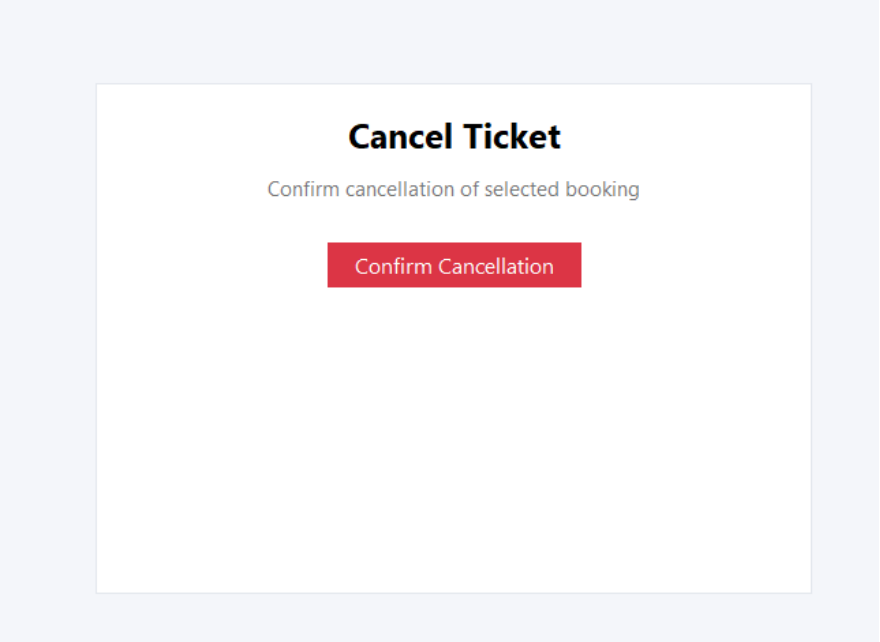

Figure 16: Cancellation Management Screen of BusYatra System

CHAPTER 7 – DEPLOYMENT & EXECUTION

The BusYatra – Online Bus Ticket Booking System can be deployed as a standalone desktop application for real-world usage. The system can be packaged into an executable file (.exe), allowing users to install and run the application directly on their computers without requiring any development environment.

This deployment approach provides the following advantages:

- Easy installation and execution on Windows systems
- No need for IDE setup such as NetBeans
- User-friendly distribution and installation
- Suitable for offline use in organizations and travel agencies

The executable file can be generated using tools such as Launch4j, which converts Java applications into Windows-compatible .exe files. This deployment method enhances the accessibility, usability, and practical implementation of the BusYatra system.

<u>Executable File (Deployment)</u>

The BusYatra application is packaged as a standalone executable (.exe) file. The executable file ensures easy distribution and installation of the system without requiring the Java Development Kit (JDK) to be separately installed.

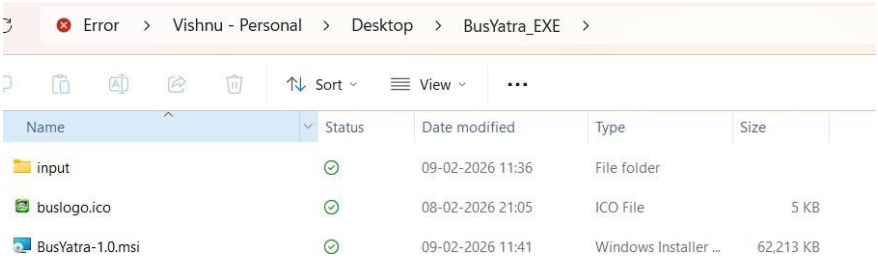

Figure 17: BusYatra Executable (.exe) File

<u>Desktop Shortcut Creation</u>

A desktop shortcut of the executable (.exe) file can be created to allow users to launch the application directly from the desktop without navigating through folders. This improves usability and provides quick access for regular use.

Figure 18: BusYatra Application Shortcut on Desktop

CHAPTER 8 – CONCLUSION

The BusYatra – Online Bus Ticket Booking System has been successfully developed to provide an efficient and organized solution for managing bus ticket reservations. The system replaces traditional manual booking methods with a computerized approach, thereby reducing errors, saving time, and improving overall efficiency.

The application offers essential functionalities such as user registration, viewing bus details, booking tickets, cancelling bookings, and managing records through a user-friendly interface. By using technologies like Java, Swing, JDBC, and MySQL (Aiven cloud), the system ensures reliable performance, secure data handling, and easy access to booking information.

Overall, the system simplifies the ticket booking process for users and reduces the workload of administrators. It also provides a scalable solution that can be further enhanced in the future by adding advanced features such as online payment integration and real-time bus tracking.

The application can also be packaged as a standalone executable (.exe) file, allowing users to download and run the system directly on their computers without requiring a separate development environment. This enhances usability and makes the system more accessible in real-world scenarios.

In conclusion, the BusYatra system fully meets the project objectives and serves as an effective tool for managing bus ticket reservations efficiently and reliably.

CHAPTER 9 – FUTURE ENHANCEMENTS

The BusYatra – Online Bus Ticket Booking System provides essential functionalities such as bus search, seat booking, ticket management, and cancellation. However, the system can be further enhanced by adding advanced features to improve its functionality, scalability, and user experience. Some possible future enhancements include:

Online Payment Integration

Integrating secure online payment methods such as UPI, debit/credit cards, and net banking for real-time transactions and instant booking confirmation.

Mobile Application

Developing a mobile application (Android/iOS) to allow users to book tickets conveniently from their smartphones, making the system more accessible.

Real-Time Bus Tracking

Adding GPS-based tracking to enable users to monitor the live location of buses and get accurate estimated arrival times.

Notification System

Implementing email/SMS notifications for booking confirmation, upcoming trip reminders, and cancellation alerts.

Advanced Seat Selection UI

Enhancing the seat layout with better visuals, color-coded availability indicators, and real-time updates for improved user experience.

Web-Based Version

Extending the system into a web-based application accessible from any browser, allowing remote access without installation.

AI-Based Recommendations

Providing smart suggestions for buses and routes based on user preferences, booking history, and travel patterns.

Improved UI/UX Design

Enhancing the overall interface with modern design elements, animations, and intuitive navigation for a better user experience.

These enhancements will make the BusYatra system more scalable, user-friendly, and suitable for large-scale real-world deployment.

CHAPTER 10 – REFERENCES

The following references were used during the development of the BusYatra – Online Bus Ticket Booking System:

1. Herbert Schildt, Java: The Complete Reference, Oracle Press, 11th Edition.
2. E. Balagurusamy, Programming with Java, McGraw Hill Education.
3. Korth, Silberschatz, Sudarshan, Database System Concepts, McGraw Hill.
4. Oracle Documentation. Available at: https://docs.oracle.com
5. Java Documentation (Oracle). Available at: https://docs.oracle.com/javase
6. JDBC Tutorial – GeeksforGeeks. Available at: https://www.geeksforgeeks.org/jdbc/
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8. W3Schools Java Tutorial. Available at: https://www.w3schools.com/java/
9. MySQL Documentation. Available at: https://dev.mysql.com/doc/
10. Aiven Documentation. Available at: https://aiven.io/docs

APPENDIX

A.1 Sample Code

Database Connection Code (Aiven MySQL using JDBC)

The following code establishes a connection between the Java application and the MySQL database hosted on Aiven using JDBC.

DBConnection.java:

```
import java.sql.Connection;
import java.sql.DriverManager;

public class DBConnection {
  public static Connection getConnection() {
    Connection con = null;
    try {
      Class.forName("com.mysql.cj.jdbc.Driver");
      con = DriverManager.getConnection(
        "jdbc:mysql://<HOST>:<PORT>/<DATABASE>?sslmode=require",
        "<USERNAME>", "<PASSWORD>");
    } catch (Exception e) {
      e.printStackTrace();
    }
    return con;
  }
}
```

InsertBooking.java:

```
import  java.sql.Connection;
import java.sql.PreparedStatement;

public class InsertBooking {
  public static void main(String[] args) {
    try {
      Connection con = DBConnection.getConnection();
      String query = "INSERT INTO bookings
          (user_id, schedule_id, total_amount, status) VALUES (?, ?,
?, ?)";
      PreparedStatement pst = con.prepareStatement(query);
      pst.setInt(1, 1);  pst.setInt(2, 101);
      pst.setDouble(3, 500.0);  pst.setString(4, "CONFIRMED");
      int i = pst.executeUpdate();
      if (i > 0) System.out.println("Booking Added Successfully");
      con.close();
    } catch (Exception e) { e.printStackTrace(); }
  }
}
```

A.2 Sample DAO Code

UserDAO.java

```
import java.sql.*;
public class UserDAO {
```

```
  public boolean registerUser(String name, String email, String
password) {
    try {
      Connection con = DBConnection.getConnection();
      String q = "INSERT INTO users (name, email, password) VALUES (?,
?, ?)";
      PreparedStatement pst = con.prepareStatement(q);
      pst.setString(1, name); pst.setString(2, email);
pst.setString(3, password);
      int i = pst.executeUpdate(); con.close(); return i > 0;
    } catch (Exception e) { e.printStackTrace(); return false; }
  }
  public boolean loginUser(String email, String password) {
    try {
      Connection con = DBConnection.getConnection();
      String q = "SELECT * FROM users WHERE email=? AND password=?";
      PreparedStatement pst = con.prepareStatement(q);
      pst.setString(1, email); pst.setString(2, password);
      ResultSet rs = pst.executeQuery();
      boolean exists = rs.next(); con.close(); return exists;
    } catch (Exception e) { e.printStackTrace(); return false; }
  }
}
```

BookingDAO.java

```
import java.sql.*;
public class BookingDAO {
  public boolean createBooking(int userId, int scheduleId, double
amount) {
    try {
      Connection con = DBConnection.getConnection();
      String q = "INSERT INTO bookings (user_id, schedule_id,
total_amount, status)
                  VALUES (?, ?, ?, ?)";
      PreparedStatement pst = con.prepareStatement(q);
      pst.setInt(1, userId); pst.setInt(2, scheduleId);
      pst.setDouble(3, amount); pst.setString(4, "CONFIRMED");
      int i = pst.executeUpdate(); con.close(); return i > 0;
    } catch (Exception e) { e.printStackTrace(); return false; }
  }
}
```

A.3 SQL Queries

Create Users Table

```
CREATE TABLE users (
  id       INT PRIMARY KEY AUTO_INCREMENT,
  name     VARCHAR(100) NOT NULL,
  email    VARCHAR(100) UNIQUE,
  password VARCHAR(100) NOT NULL
);
```

Create Buses Table

```
CREATE TABLE buses (
  id              INT PRIMARY KEY AUTO_INCREMENT,
  operator        VARCHAR(100),
  bus_type        VARCHAR(50),
  total_seats     INT NOT NULL,
```

```
  fare_multiplier DECIMAL(5,2)
);
```

Create Bookings Table

```
CREATE TABLE bookings (
  id           INT PRIMARY KEY AUTO_INCREMENT,
  user_id      INT,
  schedule_id  INT,
  total_amount DECIMAL(10,2),
  status       VARCHAR(20),
  booking_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id)
);
```

Create Booked Seats Table

```
CREATE TABLE booked_seats (
  id          INT PRIMARY KEY AUTO_INCREMENT,
  booking_id  INT,
  schedule_id INT,
  seat_id     INT,
  UNIQUE (schedule_id, seat_id),
  FOREIGN KEY (booking_id) REFERENCES bookings(id)
);
```

Create Cancellations Table

```
CREATE TABLE cancellations (
  id             INT PRIMARY KEY AUTO_INCREMENT,
  ticket_id      INT,
  user_id        INT,
  reason         VARCHAR(255),
  refund_amount  DOUBLE,
  status         VARCHAR(50),
  cancelled_at   DATETIME DEFAULT CURRENT_TIMESTAMP
);
```